

Credit Risk Scoring: PD Estimation, Threshold Optimisation, and Risk Segmentation.

1. INTRODUCTION

Every bank wants to identify people who may likely default on loans. The aim of this project is to develop a credit risk scoring model to identify which of the people may be of high risk considering the different factors and behavioural characteristics. A logistic regression model is used because it is highly interpretable and suitable for binary classification problems, where the outcome is either default or non-default.

DATA DESCRIPTION

The dataset used consists of historical loan applications and borrower information.

The target variable is `not.fully.paid`, which indicates whether a borrower defaulted (1) or fully repaid the loan (0).

The model uses financial and credit-related features including FICO score, interest rate, income (log annual income), debt-to-income ratio (DTI), credit inquiries, revolving balance and utilisation, past delinquencies, and loan purpose to estimate the probability of default.

2. DATA PREPROCESSING

The dataset did not have any missing values or inconsistencies, so the data was directly used for modelling.

3. MODEL DEVELOPMENT

Linear regression model:

$$P(Y = 1) = \frac{1}{1 + e^{-z}}, \text{ where } z \text{ is a linear combination of the input variables.}$$

The model outputs a probability between 0 and 1, which represents the likelihood that a borrower will default. Positive coefficients indicate an increase in default risk, while negative coefficients indicate a decrease in risk.

4. MODEL EVALUATION

The model was evaluated using multiple performance measures including the confusion matrix, classification report, and ROC curve.

The confusion matrix was used to assess prediction accuracy by dividing results into true positives, false positives, true negatives, and false negatives. This helps to understand how well the model identifies defaulters and non-defaulters.

The classification report provided precision, recall, and F1-score. In this context, recall for the default class is particularly important because it measures how many actual defaulters are correctly identified.

The ROC curve was used to evaluate the model's ability to distinguish between defaulters and non-defaulters across different classification thresholds.

5. THRESHOLD OPTIMISATION

Logistic regression outputs probabilities, so a classification threshold is required to convert probabilities into class labels. Instead of using only the default threshold of 0.5, different methods were tested to find an optimal threshold.

These included Youden's J statistic and a cost-based threshold approach. The final selected threshold improved the balance between false positives and false negatives, which is important in credit risk where both misclassification types have financial consequences.

6. RESULTS AND INTERPRETATION

1. Coefficients

Top positive risk factors

Feature	Coefficient
inq.last.6mths	0.153
dti	0.006
pub.rec	0.005

revol.util	0.004
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Most negative risk factors

Feature	Coefficient
fico	-0.003
credit.policy	-0.024

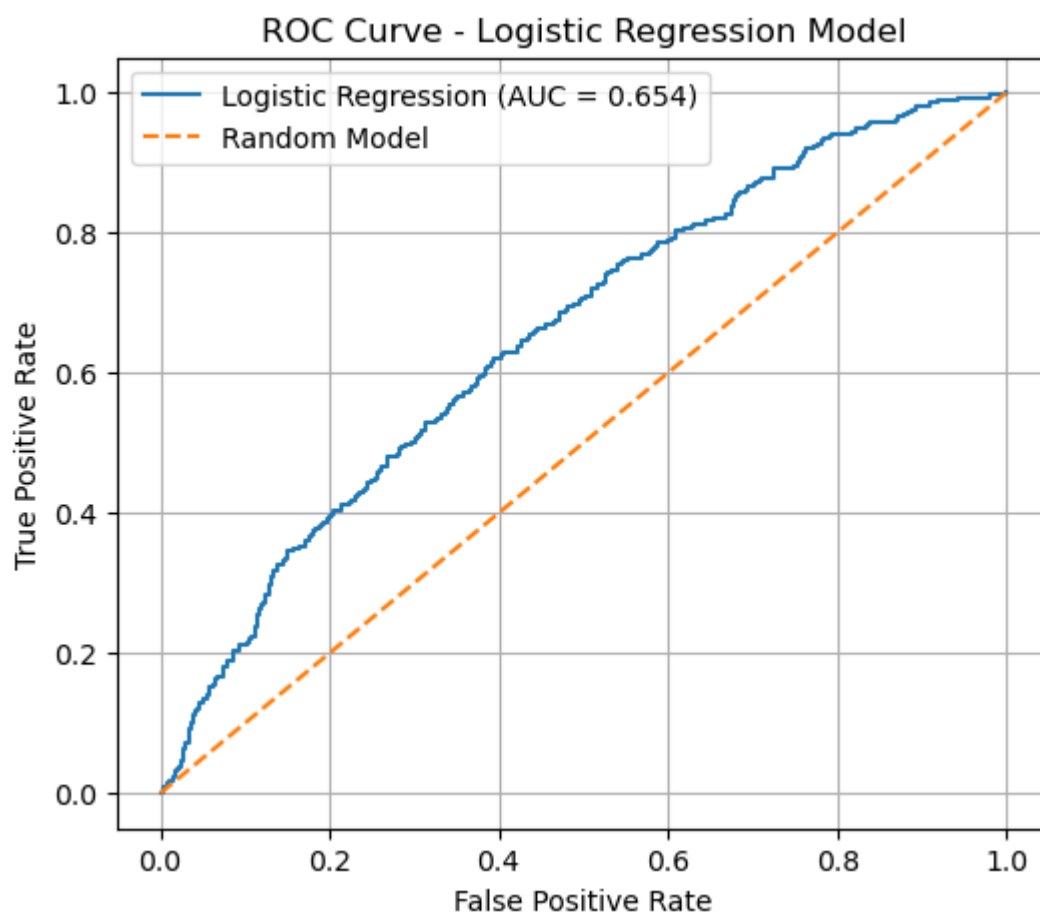
People with more recent credit inquiries, higher debt-to-income ratios and greater credit utilisation were associated with increased default risk. Higher FICO scores were associated with lower default risk.

2. Probability of default estimation(Top ten)

Rank	Observation ID	Probability of Default (PD)
1	1308	0.929987 (92.999%)
2	384	0.883326 (88.333%)
3	1576	0.670929 (67.093%)
4	829	0.652618 (65.262%)
5	1369	0.635247 (63.525%)
6	1089	0.631966 (63.197%)
7	1588	0.631821 (63.182%)
8	934	0.626034 (62.603%)
9	1259	0.614683 (61.468%)
10	1075	0.583693 (58.369%)

Table above presents the ten borrowers with the highest predicted probabilities of default generated by the model. Borrower 1308 exhibits the highest estimated probability of default (93.0%), followed by borrower 384 (88.3%). These observations represent the highest-risk segment of the portfolio and would warrant enhanced monitoring and stricter credit risk management measures.

3. ROC Curve and Discriminatory Power



The model achieved an **AUC of 0.654**, indicating a moderate ability to discriminate between good and bad borrowers. This means the model performs better than random guessing (AUC = 0.50), but there is still room for improvement in predictive power.

4. Threshold optimisation

Method	Threshold
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Default	0.50
Youden's J Statistic	0.160
Cost-Based Approach	0.072

Both Youden's J and cost-based optimisation resulted in significantly lower thresholds than the default. This reflects the importance of capturing more defaulters in credit risk applications, even at the cost of increasing false positives.

5. Final model performance

Selected threshold: 0.160

Actual / Predicted	Non-Default	Default
Non-Default	985	642
Default	111	178

The model correctly identified 178 defaulters while missing 111 actual defaults. Although a relatively large number of non-defaulters were incorrectly classified as risky, this trade-off is acceptable in credit risk settings where failing to detect a defaulter is more costly than incorrectly flagging a good borrower.

Classification report

Metric	Value
Precision	0.22
Recall	0.62
F1-score	0.32

The recall of 62% indicates that the model successfully identifies a majority of defaulters, making it useful as a risk detection tool. However, the lower precision indicates a higher number of false alarms.

Risk segmentation

PD Range	Risk Category
PD < 0.10	Low Risk
0.10 – 0.30	Medium Risk
PD > 0.30	High Risk

Borrowers were grouped into risk bands based on predicted probability of default, this allows the model outputs to be used in practical lending decisions, such as approvals, manual review, and rejections.

7. Conclusion

The model provides a moderately effective credit risk scoring system. The AUC of 0.654 shows that the model has reasonable discriminatory power, and threshold optimisation significantly improves its ability to identify high-risk borrowers compared to the default threshold.